

Techniques of High-Performance Computing - PHAS0102

Week 1: Introduction

Who am I?

- Dr. Ahmed Faris Al-Refaie
 - ahmed.al-refaie.12@ucl.ac.uk
- Principal Research Fellow in Department of Physics and Astronomy
- Research on Exoplanet Atmospheres!
- Run Advanced Python courses
- First time lecturing!

Course Admin

- Check the Moodle page!
- Forums
 - Announcements
 - Q&A
 - Asking questions about the module and content.
 - Be polite and courteous!

Course Admin: Assessment

- Coursework 1 (20%)
 - Deadline: Monday 21st October, 5 pm
- Coursework 2 (20%)
 - Deadline: Monday 4th November, 5 pm
- Coursework 3 (30%)
 - Deadline: Thursday 28th November, 5 pm
- Coursework 4 (30%)
 - Deadline: Thursday 12th December, 5 pm

Course Admin: Assessment

- Coursework is important!
 - We're only human and mistakes can happen in marking. If there you suspect a mistake, email me within a week of getting the marks back.

Getting Help

- Forums
- Lecture notes will be posted each week
- Lecturecasts will be available
- Virtual drop-in session
- Email me: ahmed.al-refaie.12@ucl.ac.uk

Announcement

- I will not be in for this Friday (4th October) lecture
 - At a conference
- Lecture will be given by Dr. Charles Bowesman
- All other Friday lectures will be by me as usual!
- Announcement will be posted in the Forum.

Course content

- HPC with Python (~ 3 weeks)
- Sparse linear algebra (~5 weeks)
- Time-dependent Problems (~2 weeks)

What is High Performance Computing?

“High-Performance Computing is concerned with developing tools, algorithms, and applications that can make optimal use of a given hardware environment.”
- Prof Timo Bette

“Computing for the cheap and impatient” – Ahmed Al-Refaie

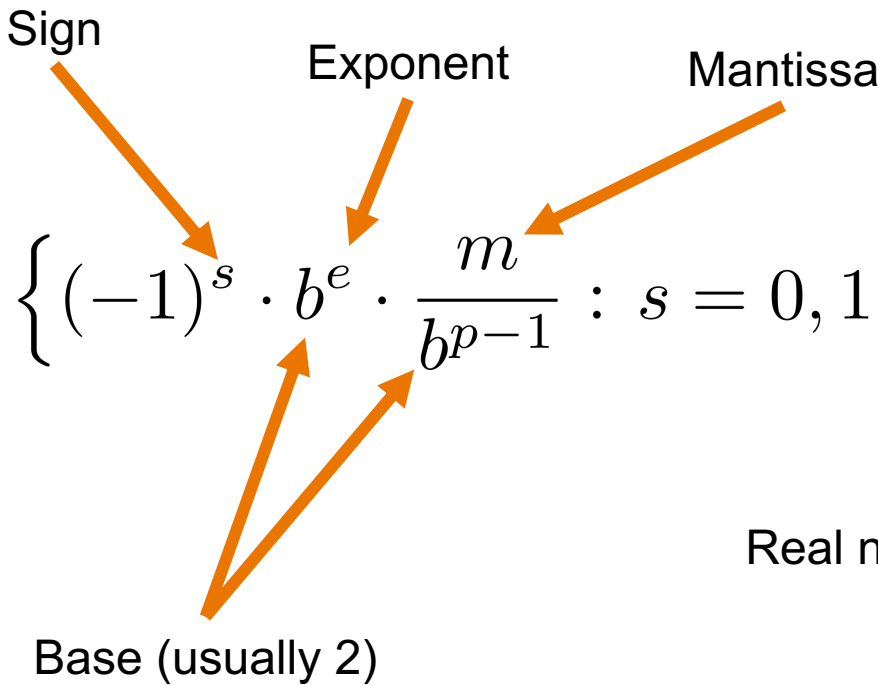
Floating point numbers

- Represent non-integer numbers in computing.
- IEEE 754 standard
- Dedicated floating point units (FPU) in hardware.
- Fixed-point alternative
 - Used in financial systems
 - Older video games used this (Starwing/Starfox, Doom)
 - Lack of FPU in old hardware.



Quake (1996) one of the first games to require an FPU

Floating point numbers



$$\mathcal{F} = \left\{ (-1)^s \cdot b^e \cdot \frac{m}{b^{p-1}} : s = 0, 1; e_{min} \leq e \leq e_{max}; b^{p-1} \leq m \leq b^p - 1 \right\}.$$

Sign

Exponent

Mantissa

Base (usually 2)

Real numbers represented by only 3 integers

Floating point numbers

$$\mathcal{F} = \left\{ (-1)^s \cdot 2^e \cdot \frac{m}{2^{p-1}} : s = 0, 1; e_{min} \leq e \leq e_{max}; 2^{p-1} \leq m \leq 2^p - 1 \right\}.$$

For the value 1:

$$s = 0, e = 0, m = 2^{p-1} \rightarrow 1$$

For the next value after 1:

$$s = 0, e = 0, m = 2^{p-1} + 1 \rightarrow 1 + 2^{1-p}$$

Floating point numbers

- Floating point numbers are not equally spaced
- Consider the mantissa term $\frac{m}{2^{p-1}}$ for numbers between 1 and 2

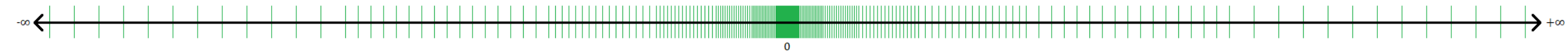
$$\frac{m}{2^{p-1}} = 1, 1 + 2^{1-p}, 1 + 2 \times 2^{1-p}, \dots, 2 - 2^{1-p}.$$

- If we move to $e = 1$ then we will have mantissa representing 2 to 4

$$2^1 \frac{m}{2^{p-1}} = 2, 2 + 2^{2-p}, 2 + 2 \times 2^{2-p}, \dots, 4 - 2^{2-p}.$$

- And so on ... the spacing getting coarser

Floating point numbers



Floating point

$$\mathcal{F} = \left\{ (-1)^s \cdot 2^e \cdot \frac{m}{2^{p-1}} : s = 0, 1; e_{min} \leq e \leq e_{max}; 2^{p-1} \leq m \leq 2^p - 1 \right\}.$$

Single precision ('float', 'float32')

- $e_{min} = -126$ $e_{max} = 127$
- $p = 24$

Storage

- Exponent: 8 bit
- Mantissa: 23 bits
- Sign: 1 bit
- Total: 32 bits

Double precision ('double', 'float64')

- $e_{min} = -1022$ $e_{max} = 1023$
- $p = 53$

Storage

- Exponent: 11 bit
- Mantissa: 52 bits
- Sign: 1 bit
- Total: 64 bits

Floating Point numbers

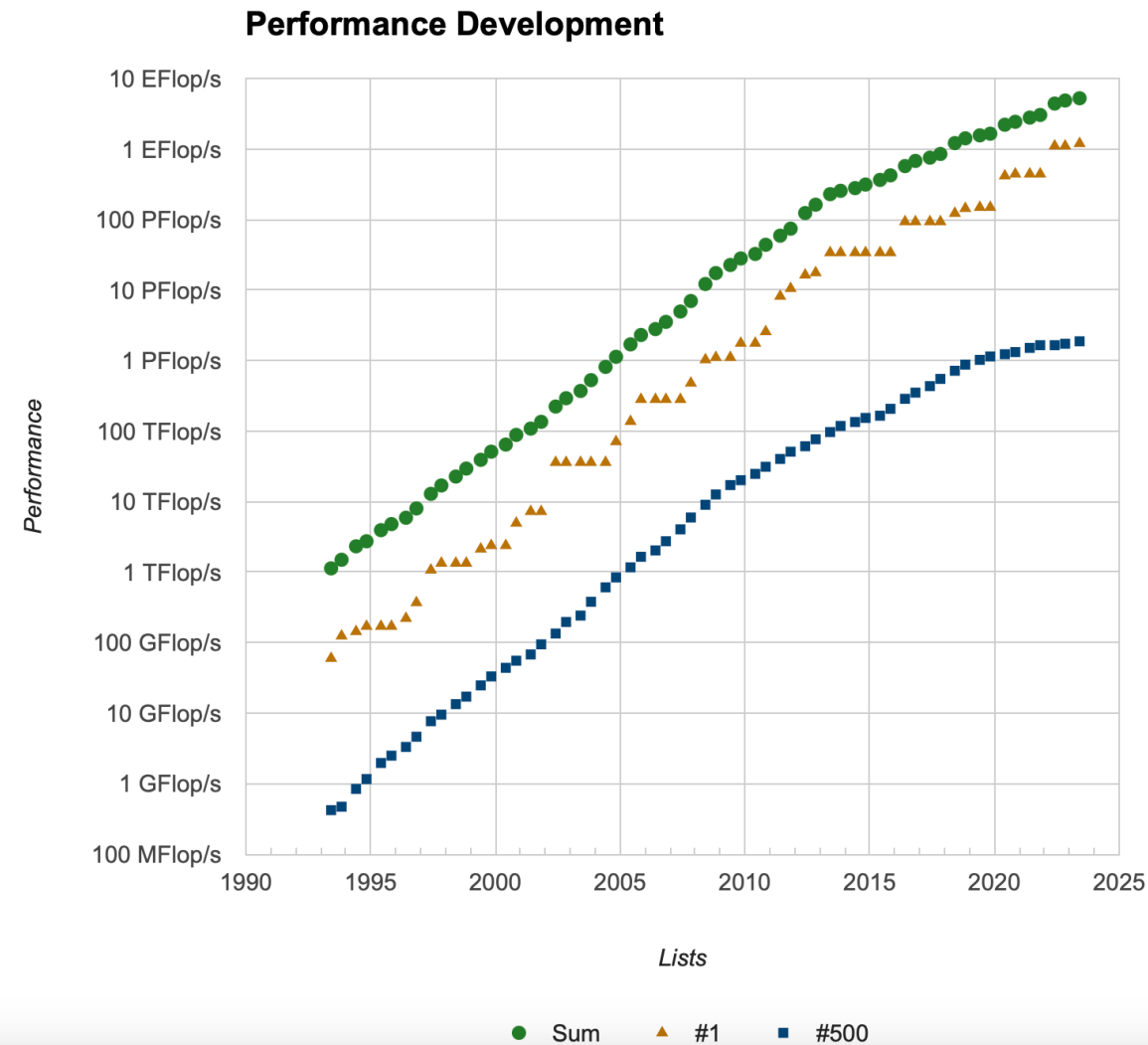
- ε_{rel} is the smallest possible value such that $1 + \varepsilon_{rel} \neq 1$
- Single precision (32 bits):
 - $\varepsilon_{rel} \approx 1.2 \times 10^{-7}$
- Double precision (64 bits):
 - $\varepsilon_{rel} \approx 2.2 \times 10^{-16}$

FLOP/s

Floating Point **O**perations per second

Machine	Peak Performance (GFlops/s)	
Intel Xeon Platinum 8280M (28 Cores)	1,612	A fast workstation CPU
Raspberry PI 4 Model B	24	Cheap and Fun ARM CPU
Nvidia RTX 4090	73,000	FP32 Peak
PS5 GPU	10,280	FP32 Peak
XBOX Series X GPU	12,500	FP32 Peak
PS5 Pro GPU	36,000	FP32 Peak

FLOP/s



High Performance Computing with Python

- Google Colab (*Recommended*)
- Anaconda / Miniconda